Multilevel Attention Siamese Network for Keypoint Detection in Optical and SAR Images

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***Abstract*— Optical and synthetic aperture radar (SAR) image keypoint detection is an important foundation for multimodal remote sensing image matching. The influence of nonlinear radiometric differences and geometric deformation between optical and SAR images leads to low repeatability of existing keypoint detection methods. To address the problem that existing keypoint detection methods cannot provide the required homonymous points for heterogeneous image matching, we propose a keypoint detection method [multilevel attention Siamese network for keypoint detection in optical and SAR images (SKD-Net)] and improve it in terms of both network structure and network optimization. First, we propose a multilevel attention Siamese network, which is composed of multiple convolutional modules and transformer modules with shared weights to extract common features at different levels for keypoint detection. We introduce a transformer module in the keypoint detection pipeline and fuse shallow and deep features to obtain more spatial and rich semantic information to facilitate heterogeneous image keypoint detection. Then, to ensure that the detected keypoints have more homonymous points and localization accuracy, we propose a position-consistent loss. Unlike previous loss functions, our designed position-consistent loss function takes the differences between heterogeneous image score maps into account, and it autonomously selects the optimized correct point pairs to enable the network to perform correct learning. Finally, extensive experiments show that our detection method outperforms the current state-of-the-art keypoint detection methods in terms of repeatability, localization accuracy, and matching performance. Our source code is available at https://github.com/zhangschen/SKD-Net.**

***Index Terms*— Deep learning, keypoint detection, optical and synthetic aperture radar (SAR) images.**

I. Introduction

With the rapid development of sensor technology, space technology, and communication technology, modern remote sensing technology has entered the "multisource era" (multisensor, multispectral, and multitemporal). With the advantage of complementary multimodal image information, multimodal images have been widely used in image fusion [1], [2], change detection [3], [4], image matching [5], [6], and other fields. Synthetic aperture radar (SAR) has various features, such as all-day and all-weather imaging capabilities and cloud penetration ability, yet it has problems such as coherent speckle noise, which makes the images difficult to interpret. In contrast, optical images can reflect the spectral and spatial information of features and are easy to interpret, but they are easily interfered with by clouds and fog, resulting in information loss. Matching optical and SAR images and performing image fusion can realize information complementation between different types of sensor imaging, which can better facilitate subsequent image analysis and interpretation. Therefore, optical and SAR image matching is one of the core problems that urgently needs to be solved to realize the above operations.

Many scholars have started to study the optical and SAR image registration; for example, Xiang et al. [7] proposed a fast extraction of deep and shallow feature descriptors based on cross-stage partial Siamese networks. Xiang et al. [8] proposed a feature decoupling network-based optical and SAR image registration method to overcome the shortcomings of feature- and region-based accuracy and computational power. Zhou et al. [9] constructed multiscale convolutional gradient features by multiscale convolution of gradient features through shallow pseudo-Siamese networks to refine structural features to improve image registration. Ye et al. [10] constructed a channel-controllable filter structure descriptor by combining first and second-order gradient information with a multiscale strategy. Although they achieved better results in optical and SAR image registration, they put more emphasis on feature extraction of the descriptors and little focus on the keypoint detection part. For optical and SAR image registration, we need not only robust descriptors but also reproducible keypoints, and our work is to provide more reproducible keypoints for optical and SAR image matching.

Due to the different imaging mechanisms and working methods, there are significant geometric deformation and nonlinear radiation differences between optical and SAR images [11], which make it difficult for existing keypoint detection methods to obtain a sufficient number of homonymous points for optical and SAR image matching to meet the demand of high-precision matching. For this reason, several

Manuscript received 21 April 2023; revised 29 June 2023; accepted 4 July 2023. Date of publication 7 July 2023; date of current version 19 July 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 41961053 and in part by the Yunnan Fundamental Research Projects under Grant 202301AT070463 and Grant 202101AT070102. *(Corresponding author: Zhitao Fu.)*

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Digital Object Identifier 10.1109/TGRS.2023.3293143

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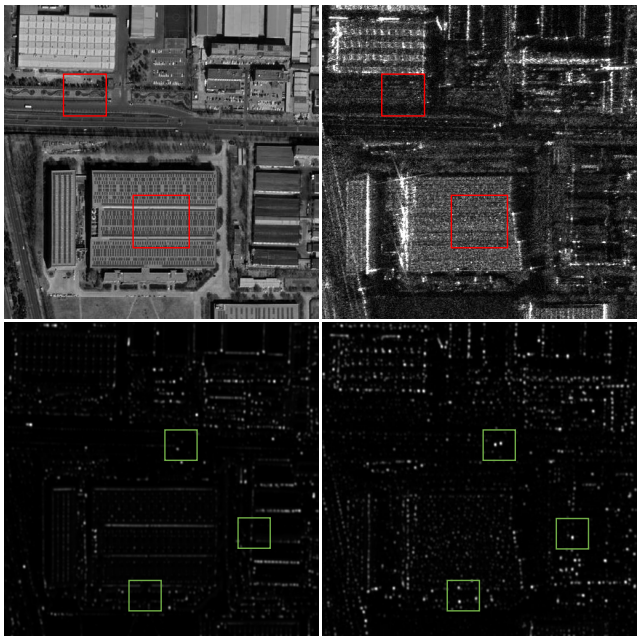


Fig. 1. Pair of optical and SAR images and their corresponding score maps. The first row shows the optical and SAR images, and the second row shows the score maps corresponding to the optical and SAR images.

challenges for optical and SAR image keypoint detection remain unresolved. First, the nonlinear radiometric differences and geometric deformation between heterogeneous images lead to different interpixel correlations, different statistical properties, and large texture differences between heterogeneous images. As shown in Fig. 1, there are significant appearance differences between optical and SAR images acquired for the same area and in the red box due to the lack of texture features in SAR images, resulting in large differences in extracting local features at the same location. Second, SAR images contain a large amount of scatter noise, which will interfere with local feature learning, as shown in the green box in Fig. 1, with different score values on the optical and SAR score maps at the same locations, leading to difficulties in detecting points at the same locations on heterogeneous images. Therefore, the main challenge of heterogeneous image keypoint detection is how to take the differences between heterogeneous images and extract local common features that are invariant to the modal transformation of the images into account so that the constructed score maps have the same score values at the same locations.

Traditionally, keypoint detectors are mainly based on handcrafted detectors, and handcrafted keypoint detectors [12], [13], [14], [15] are more biased toward finding corner points or speckles in images. In multimodal remote sensing image applications, these handcrafted detectors are often used directly or improved to accommodate the variability of multimodal images. Histogram of oriented phase congruency (HOPC) [11] extracts keypoints with a Harris [12] detector and then describes the acquired keypoints for template matching. The detection part in radiation-variation insensitive feature transform (RIFT) [16], on the other hand, uses the minimum and maximum moment maps. The features from accelerated segment test (FAST) [17] detector detects corner point features on a minimum moment map, which eliminates the effect of certain nonlinear radiometric differences by extracting common features between heterogeneous images. Fan et al. [18] performed multiscale feature point detection by combining nonlinear diffusion and spatially consistent strategies into the Harris–Laplace detector. In [19] and [20], on the other hand, the scale invariant feature transform (SIFT) [14] detector was improved, and although the performance in heterogeneous image matching was improved to some extent, the keypoint detection part did not achieve good results.

With the development of deep learning in recent years, an increasing number of researchers have proposed keypoint detectors in deep learning [21], [22], [23]. Deep learning methods acquire the features of an image with convolutional neural networks (CNNs), construct the corresponding response graph or score graph, and use the local maximum score value of the score graph as the keypoint. Existing deep learning keypoint detection methods can be divided into two main categories: detection and description of joint learning [24], [25], [26], [27], [28] and detector-only learning [29], [30], [31]. With deep learning methods achieving significant advantages in global feature learning and local feature learning [32], [33], image matching is increasingly focused on jointly learned detection and description. In [24] and [26], jointly trained networks for keypoint detection and networks for local feature description were used with deep network architectures. To enhance the connection between feature detection and description to improve the reliability of detected features, Revaud et al. [27] and Dusmanu et al. [34] performed both keypoint detection and obtained dense feature descriptions through a single CNN. Although joint detection and description methods achieve good results in terms of matching, they focus more on the description part, the extracted keypoints are not localized with high accuracy, and the keypoint detection is strongly influenced by the descriptors. Therefore, some methods [29], [30], [31], [35] focus only on the keypoint detection part. Temporally invariant learned detector (TILDE) [30] identifies keypoints by training multiple segmented linear regression models, which have better robustness under weather and lighting drastic imaging transformation conditions. To obtain more robust keypoints, Key.Net [29] attempts to combine manual and learned CNN filters for keypoint detection in a shallow multiscale architecture to detect robust features present at different scales.

Although the above methods have achieved better results in homologous images, the large differences between optical and SAR images make them often have problems such as low repetition rate and poor keypoint localization accuracy when performing keypoint detection in heterologous images. At present, scholars have paid attention to the above problems and proposed some methods applicable to heterogeneous image keypoint detection. For example, Li et al. [36] detected repeatable keypoints in heterogeneous images by computing the local space and channel domain peaks of a feature map. ReDFeat [37] combines the detection and description constraints of feature learning through a weighting strategy to peak and repeat the detected robust features to meet the demanding detection conditions, and although it achieves better results in heterogeneous image matching, the keypoints of detection tend to appear in texture-free regions due to the influence of descriptors.

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To address the problem that existing keypoint detection methods do not provide the required number of homonymous points for heterogeneous image matching, our method aims to provide more repeatable keypoints for heterogeneous image matching. We improve our keypoint detector from two aspects, namely, network structure and network optimization. In terms of network structure, to overcome cross-modal intensity variations and obtain detailed local common information, we design a multilevel attention Siamese network with two convolution modules and a transformer module inside the network to extract features at different levels. To extract more spatial and rich semantic information for keypoint detection, we introduce a transformer into the keypoint detection pipeline. Since shallow features contain more basic information and are more robust to geometric transformations and deep features contain more common information, we fuse shallow and deep information to obtain different levels of information. In terms of network optimization, we design a new position-consistent loss applicable to both optical and SAR image detection. Specifically, we design a position consistency loss with autonomous selectivity, which autonomously chooses to optimize the correct keypoint pairs to enable the network to perform correct learning. This accommodates the variability between heterogeneous image score maps and guarantees the repeatability and localization accuracy of keypoints.

The contributions of this article are summarized as follows.

1) In this article, we propose a multilevel attention Siamese network. We introduce a transformer into the keypoint detection pipeline and fuse shallow and deep features to obtain more detailed local commonality information for keypoint detection.
2) We design a position-consistent loss function for keypoint detection of heterogeneous images, taking the differences between heterogeneous image score maps into account.

The rest of this article is organized as follows. Section II describes the work related to image keypoint detection methods. Section III presents the details of our proposed method. Section IV presents the experimental results and analysis. Section V draws the conclusion.

II. Related Work

This section introduces traditional keypoint detection methods and deep learning keypoint detection methods. We present the deep learning keypoint detection methods in two parts: detector-only keypoint detection methods and joint detection and description keypoint detection methods.

A. Traditional Keypoint Detection Methods

Traditional keypoint detection methods rely heavily on handcrafted keypoint detectors that find corner points or speckles in an image by algorithmically locating the geometric structure. For example, Harris and Stephens [12] found corner points in an image by using first-order derivatives. SIFT [14] uses the Gaussian difference function as the response operator to find the keypoints of an image, which are robust to changes in image scale, brightness, and rotation. To reduce boundary blurring and detail loss, KAZE features [38] finds the keypoints of an image by constructing a nonlinear scale space and finding the keypoints in the nonlinear scale space. However, they are computationally time-consuming, and to improve the computational speed of keypoint detection, speeded up robust features (SURF) [15] uses Harr filters and integral images to reduce the computational requirements. To cope with the nonlinear radiometric differences between multimodal images, PC-FAST [16] performs FAST [17] keypoint detection on a maximum moment map by constructing maximum and minimum moment maps with local maximum detection and nonmaximum suppression on a minimum moment map. However, the robustness and performance of manually designed keypoint detection methods for images with large nonlinear radiometric disparity need further improvement.

B. Keypoint Detection Method Based Solely on Detectors

In recent years, with the success of neural networks in local feature extraction for computer vision, an increasing number of researchers have started to experiment with their applications in keypoint detection. For example, D2D [39] uses descriptors with high information content in the descriptor space that are distinct from local descriptors as keypoints by applying saliency measures to the descriptors. To obtain keypoints that are robust under drastic weather and lighting imaging transformation conditions, TILDE [30] identifies keypoints by training multiple segmented linear regression models. Li et al. [36] proposed a self-supervised Siamese network to detect keypoints by computing the local spatial domain and channel domain peaks of the feature map. Since low-level features are rich in shape, line, and corner information, some researchers have attempted to use convolutional networks to obtain shallow-level features of images for keypoint detection. For example, KCNN [31] acquires shallow features of an image for keypoint detection by using a three-layer convolutional network. To obtain more robust keypoints, Key.Net [29] attempts to combine manual and learned CNN filters for keypoint detection in a shallow multiscale architecture to detect robust features present at different scales. Since shallow features contain more basic information and are more robust to geometric transformations, this inspired us to use shallow information from heterogeneous images for keypoint detection.

C. Keypoint Detection Method for Joint Detection and Description

As deep learning achieves significant advantages in global and local feature learning, image matching has become more focused on joint learning for detection and description. For example, learned invariant feature transform (LIFT) [24] introduces a deep network architecture for integrated learning of feature detection, feature description, and orientation estimation. To obtain scale-invariant keypoints and ensure the repeatability of keypoints, LF-Net [25] proposes a novel deep architecture for learning local features, jointly training a keypoint detection network and a local feature description network. Considering that multilevel discriminative features

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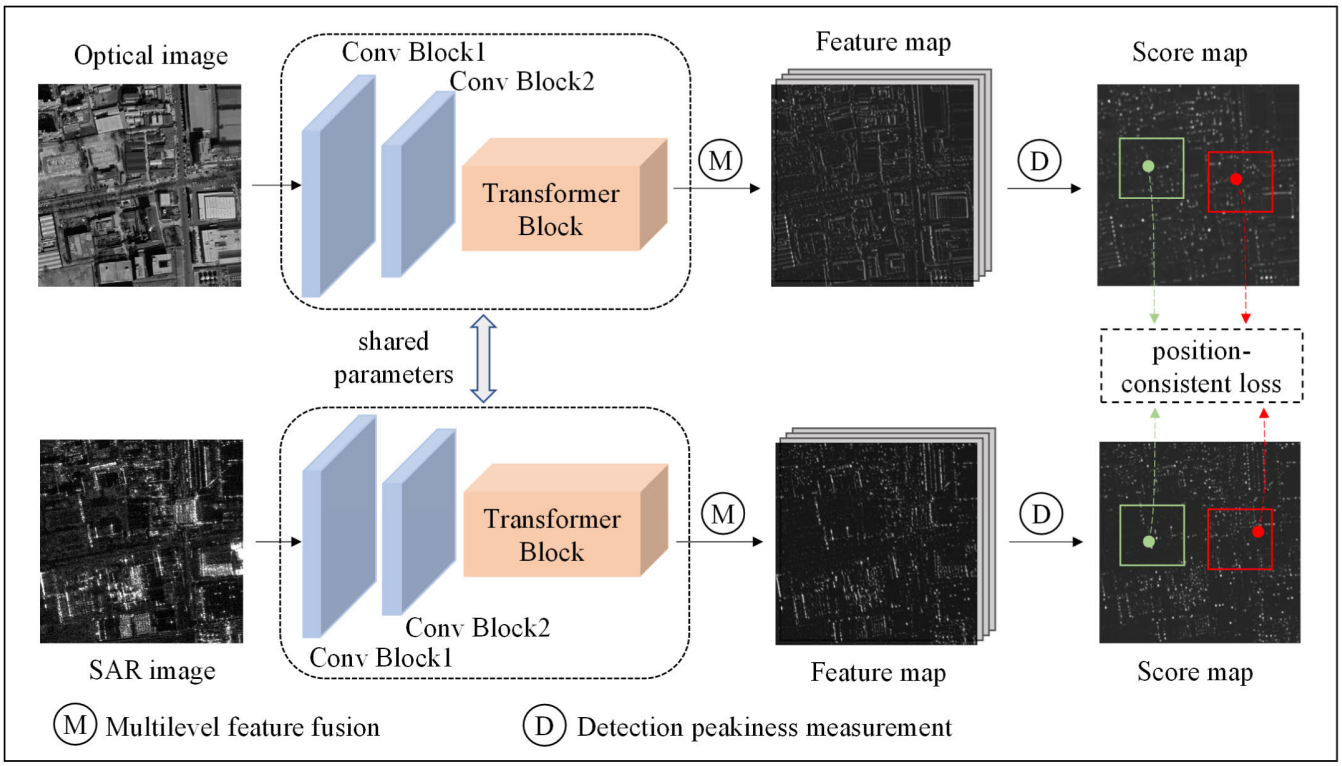


Fig. 2. Schematic of the proposed multilevel attention Siamese network for optical and SAR image keypoint detection. The green points in the figure indicate the correct point pairs, which are involved in the loss calculation after passing the condition constraint. The red points indicate the incorrect point pairs, which do not participate in the loss calculation after passing the condition constraint.

contain different sampling information, RF-Net [26] constructs response maps by extracting features at different levels of a neural network to obtain repeatable keypoints. In these methods of detection followed by description, there is an obvious limitation that detection and description cannot reinforce each other. To strengthen the connection between them, SuperPoint [28] uses the same-origin supervised domain adaptation framework for keypoint detection and description, in which keypoint detection is trained by synthesizing shape datasets. To further strengthen the connection between feature detection and description to improve the reliability of detected features, R2D2 [27] and D2-Net [34] perform both keypoint detection and dense feature description through CNNs. R2D2 [27] maximizes the detection fraction locally by relaxing the constraints, which allows the detection of more sparse, repeatable, and reliable keypoints. D2-Net [34] optimizes both detection and description by suppressing the detection fraction of descriptors and guides them to each other, obtaining points with stronger robustness under difficult imaging conditions. The keypoint detection method of joint detection and description by extracting deep common features can eliminate the variability between images to a certain extent and is more conducive to matching. While this approach seems appealing, it focuses more on the description part, and the keypoint detection part is strongly influenced by the descriptors, which often leads to a poor repetition rate and low localization accuracy in keypoint detection. The extraction of deep common feature maps can eliminate the variability between images to some extent; thus, we attempt to take advantage of the deep features to perform keypoint detection.

III. Method Introduction

In this article, we propose a multilevel attention Siamese network for optical and SAR image keypoint detection, and the whole network framework is shown in Fig. 2. First, the optical and SAR images are mapped to the shared feature space using a convolution module of shared feature mapping and a transformer module. Second, multilevel feature fusion is performed on the shared shallow features and deep layers. Third, the fused features are aggregated by peak measurement into a keypoint detection final score map. During the training process, we use the position-consistent loss to optimize the network. The details of the multilevel attention Siamese network structure and optimization method are described in Section III-A.

A. Multilevel Attention Siamese Network Structure

In computer vision applications, keypoint detection focuses more on local image features, which requires a good keypoint detection model to focus more on local feature changes of an image to achieve pixel-level detection. There are significant geometric distortion and nonlinear radiometric differences between optical and SAR images, which destroy the pixel change consistency of the pixel changes of these images in the same region. Thus, the keypoint detection of optical and SAR images has a low repetition rate and low localization accuracy. To solve this challenge, extracting local similarity features of optical and SAR images has become an important breakthrough. We have designed a Siamese network structure inspired by the successful use of Siamese networks in

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heterogeneous image matching [40], [41], [42]. A weight-sharing Siamese network can overcome cross-modal intensity variations and extract common features. Considering that multiple pooling in traditional CNNs loses too much local transform information and that pure convolutional networks are inherently limited by the size of the receptive field per pixel, many recent efforts have focused on introducing attentional mechanisms [43], [44], [45] to further increase the receptive field of the model. Since transformers have demonstrated their powerful feature extraction capability [46], we introduce a powerful transformer attention mechanism to the keypoint detection pipeline to obtain a more robust keypoint detection network. The detailed structure of our proposed multilevel attention twin network is given as follows.

1) Multimodal Feature Learning Convolution Module: To adaptively learn similarity features from optical and SAR images that differ greatly in appearance and extract features at different levels of optical and SAR images, we use two convolutional modules for feature extraction. Assuming that an optical image is represented by f_0 in a multimodal image, the learned features of the multimodal image can be expressed as follows:

$$f_1 = C_1(f_0)$$
$$f_2 = C_2(f_1) \tag{1}$$

where C_1 denotes the first convolutional module in multimodal feature learning, C_2 denotes the second convolutional module in multimodal feature learning, f_1 is the feature learned by f_0 under the first convolutional module, and f_2 is the feature learned by f_1 under the second convolutional module.

The network structures of C_1 and C_2 are shown in Fig. 3. C_1 contains two convolutional layers, each followed by a batch normalization (BN) operator and a sigmoid activation function, with a convolutional kernel of size 3×3 and a convolutional filter of 32. The convolutional structure of C_2 is different from that of C_1, and we use a basic block structure similar to that of ResNet. Maximum pooling is used before the convolution, and the maximum pooling downsampling rate is 1/2. Each convolution is followed by a BN operator, a sigmoid activation function is used, the size of the convolution kernel is 3×3, and the number of convolution filters is 64.

2) Transformer Module: Recently, transformers have received increasing attention in computer vision, and transformer-based models have achieved large performance gains in computer vision tasks due to their powerful feature extraction capability. For example, in remote sensing image transform detection, Chen et al. [47] proposed a dual-temporal image transformer to model the context in the spatiotemporal domain, extracting the transformed regions of remote sensing images. In remote sensing image classification, Yu et al. [48] proposed a multilevel spectral–spatial transformer network to perform hyperspectral image classification by using a transformer to mine multiple time series patterns. In remote sensing image matching, Nie et al. [45] used a transformer attention mechanism to extract features of weak texture regions and proposed a multiscale unsupervised orientation estimation method to achieve high-precision rotation invariant matching of remote sensing images. In this article, we explore the potential of a transformer in the keypoint detection of heterogeneous remote sensing images, and we merge a transformer module into the keypoint detection pipeline because we want to take advantage of convolution and transformer to extract features that facilitate the keypoint detection of optical and SAR images. To reduce the computational effort of learning the attention mechanism of the transformer, we use a convolution block to downsample f_2 and the channel transform. The transformed feature f_2' is represented as follows:

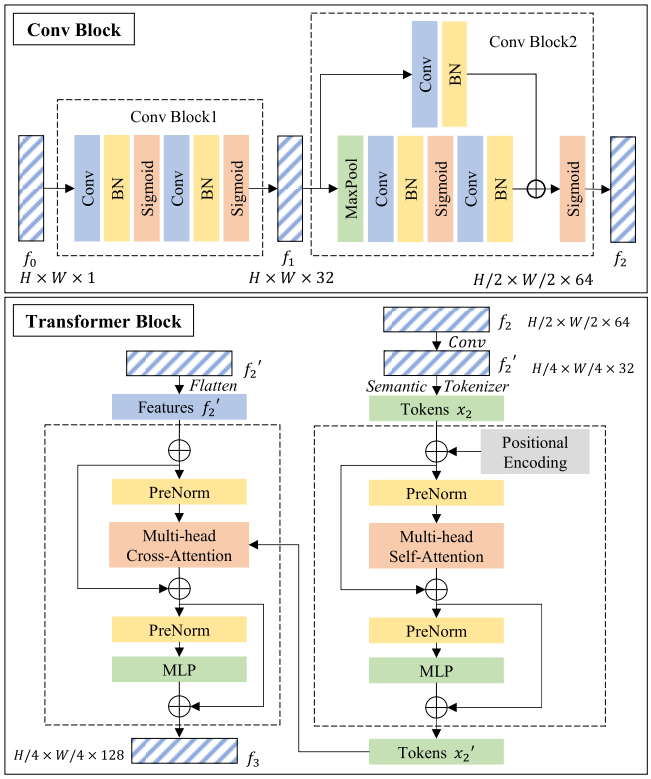


Fig. 3. Schematics of the convolutional and transformer modules in the proposed multilevel attention Siamese network.

$$f_2' = C_2'(f_2) \tag{2}$$

where C_2' is the feature transform network of f_2, consisting of maximum pooling with a downsampling rate of 1/2 and a layer of convolution. The size of the convolution kernel is 3×3, and the number of convolution filters is 32.

In transform detection, Chen et al. [47] argued that changes of interest in images can usually be expressed in terms of several high-level concepts, i.e., semantic markers. Since corner points and speckles are usually point-of-interest transformations in images, we argue that in keypoint detection, point-of-interest transformations can also be expressed in terms of several high-level concepts, and for this purpose, we use a semantic tagger to obtain compact semantic tags from optical and SAR feature maps. The semantic tagger splits the whole image into several visual words, each corresponding to a labeled feature vector. We set the length of the set of words used for tagging to L. The multimodal features f_2' are semantically tagged into a semantic token set x_2 of length L,

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which is calculated as follows:

$$x_2 = \left(\zeta\left(\varepsilon\left(f_2', \theta\right)\right)\right)^T f_2' \tag{3}$$

where $\zeta(\cdot)$ denotes the softmax activation function, $\varepsilon(\cdot)$ denotes the point-by-point convolution operation, and θ is the number of convolution filters, whose number is the length L of the set of labeled words. We will discuss the value of L in the ablation experiments in Section IV.

a) Transformer encoder (TE): After obtaining the semantic token set x_2 with multimodal features through semantic tagging, we use a TE [49] to model these semantic token sets in context. As shown in Fig. 3, we input the semantic token set x_2 into the TE to obtain a new semantic token set x_2'.

Before feeding the semantic token set x_2 into the TE, we need to add the location encoding to the semantic token set and feed it to the transformer based on the information about the relative or absolute location of the word set, which is more beneficial for context modeling. For example, the information of local transformations can guide the transformer to use the information of the transformations to correlate the contexts. In the TE, there are multilayer multiheaded self-attention (MSA) and multilayer perceptron (MLP) components.

MSA has the advantage of joint processing of subspace information with different representations at different locations. MSA consists of multiple self-attention blocks that are connected independently of each other. For each self-attention layer, the inputs are the query Q, the key K, and the value V. They are calculated as follows:

$$\begin{aligned} Q &= x_2 W^q \\ K &= x_2 W^k \\ V &= x_2 W^v \end{aligned} \tag{4}$$

where x_2 is the input semantic token set, $W^q \in U^{C\times d}$, $W^k \in U^{C\times d}$, and $W^v \in U^{C\times d}$ are learnable parameters of three linear projection layers, c is the number of multimodal feature channels, and d is the channel dimension of the triple projection layers. The formula for calculating a single attention value is given as follows:

$$\text{Attention}(Q, K, V) = \zeta\left(QK^T \Big/ \sqrt{d}\right)V \tag{5}$$

where $\zeta(\cdot)$ denotes the softmax activation function in the channel dimension. The multiheaded attention value is calculated as follows:

$$\text{MSA}(Q, K, V) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \tag{6}$$

where $W^O \in U^{C\times hd}$ denotes the linear projection matrix and h denotes the number of attention heads. head_1 is calculated as follows:

$$\text{head}_i = \text{Attention}\left(QW_i^q, KW_i^k, VW_i^v\right) \tag{7}$$

where W_i^q, W_i^k, and $W_i^v \in U^{C\times d}$ denote the linear projection matrix.

In the MLP by two linear transformation layers, the MLP is activated by a Gaussian error linear unit (GELU). Its MLP is calculated as follows:

$$\text{MLP}(x_2) = \text{GELU}(x_2 W_1)W_2 \tag{8}$$

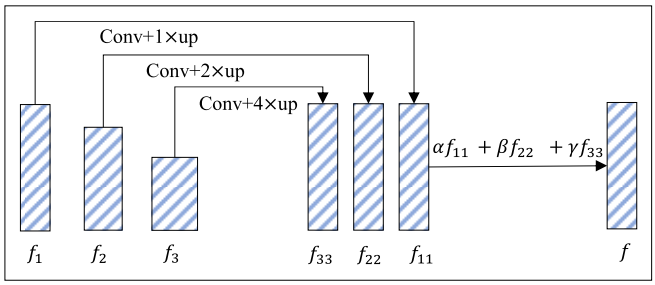


Fig. 4. Multilevel feature fusion.

where W_1 and W_2 are linear projection transformation matrices.

b) Transform decoder (TD): After encoding is performed by the transformer, we obtain a new set of semantic tokens x_2'. These context-rich semantic token sets contain compact high-level semantic information and can represent changes of interest very well. For rich pixel-level features, we need the TD to project the semantic token set x_2' back into the pixel space. Unlike in the TE, our TD consists of multiple cross-attention (MA) and MLP blocks, where the MLP block is calculated as in the TE, and the MA is calculated as follows:

$$\text{MA}\left(x_2, x_2'\right) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)W^O \tag{9}$$

where $W^O \in U^{C\times hd}$ denotes the linear projection matrix and h denotes the number of attention heads. head_1 is calculated as follows:

$$\text{head}_i = \text{Attention}\left(x_2 W_i^q, x_2' W_i^k, x_2' W_i^v\right) \tag{10}$$

where W_i^q, W_i^k, and $W_i^v \in U^{C\times d}$ denote the linear projection matrix.

3) Multilevel Feature Fusion: In homologous keypoint detection, low-level features represent the shape, line, and corner information in a scene that is robust to geometric transformations [31], [50]. To acquire low-level features, Suwanwimolkul et al. [51] used a shallow network to acquire keypoints, and Barroso-Laguna and Mikolajczyk [29] and Di Febbo et al. [31] acquired keypoints through a combination of manual and shallow networks. Due to the nonlinear radiometric differences and significant geometric deformation between optical and SAR images, there is a large difference between optical and SAR image features when acquiring shallow features, and more common information cannot be obtained. When deep feature extraction is performed, more common information between heterogeneous images can be obtained, but their local detail information is lost, and more shape, line, and corner information in the scene cannot be obtained. To solve these problems, we extract features at different levels in the Siamese network for fusion, and the obtained feature maps contain richer semantic information, which is more conducive to heterogeneous image keypoint detection.

Our proposed multilevel feature fusion approach is shown in Fig. 4. f_1, f_2, and f_3 are the feature maps obtained by the convolution module and transformer module. We map f_1, f_2, and f_3 to a convolution with a convolution kernel size

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of 1×1 and a convolution filter of 32 for channel variation and then upsample them by bilinear interpolation to obtain feature maps f_{11}, f_{22}, and f_{33} with the same channel size. We assign different weighting coefficients to different levels of features for fusion to obtain the final feature map f. The formula is given as follows:

$$f = \alpha f_{11} + \beta f_{22} + \gamma f_{33} \tag{11}$$

where α, β, and γ are weighting coefficients that sum to one. We will discuss the effect of their different weighting coefficients on the detection in the ablation experiments in Section IV.

4) Keypoint Peakiness Measurement: In the keypoint detection score map construction process, most detection methods [28], [29], [52] often use convolution to construct the score map by transforming the multichannel feature map into a single-channel feature response map. In contrast, in joint detection and described keypoint detection methods [53], [54], a layer of the multichannel feature map is used as the score map for keypoint detection. The score maps constructed in this way do not correlate feature responses well enough to derive more indicative detection scores [50]. In [34], [36], and [50] better detection results were achieved by using channel domain and spatial domain information to construct indicative detection scores. Therefore, we use the peak measure in [36] to construct the score map, which is derived from the local score and the channel score. The local spatial score for each position (i, j) in the feature map $f^c \in R^{H\times W\times C}(c = 1, 2, 3, \ldots, C)$ is calculated as follows:

$$\alpha_{ij}^c = \text{softplus}\left(f_{ij}^c - \frac{1}{|\text{N}(i,j)|}\sum_{(i',j')\in \text{N}(i,j)} f_{i'j'}^c\right) \tag{12}$$

where $\text{N}(i, j)$ denotes the set of nine neighboring pixels of pixel i and j and its own pixel value. c denotes the first few channels, f_{ij}^c denotes the value of (i, j) on the feature map of the cth channel, and softplus is the activation function. The channel score is calculated as follows:

$$\beta_{ij}^c = \text{softplus}\left(f_{ij}^c - \frac{1}{c}\sum_t f_{ij}^t\right) \tag{13}$$

where c denotes the first few channels, and the score map is constructed by the product of the channel score and the local space score with the following equation:

$$\gamma_{ij} = \max_c \left(\alpha_{ij}^c \beta_{ij}^c\right). \tag{14}$$

B. Position Consistency Loss

Existing keypoint detection losses can be divided into two main categories: constraint by score value and constraint by position. For example, Ono et al. [25], Shen et al. [26], and Quan et al. [54] constrained the loss by constraining the score values on the feature response maps so that the scores at the same locations are consistent. Barroso-Laguna and Mikolajczyk [29] and Quan et al. [54] constrained the location of the local maximum score points so that the local maximum score points at the same locations on the two score maps are in the same position. Although these approaches achieve better results with homogenous images, errors occur when applying them to heterogeneous images; for example, the positions of a large number of detected points appear in regions where there are no features. The main reason for this situation is that there is a large difference in the score maps between homologous and heterologous images, and the loss function optimizes the score points without a pair of homonyms, causing the network to perform incorrect learning.

As shown in Fig. 5(a), in the homologous image when the local range is 8×8 and 16×16, the local maxima of the two score maps are basically at the same position, and they have the same maximum score value in the same local range. This makes the loss function in the homologous source, whether by the position constraint or the score value constraint, allow the same score value at the same position and enables the network to perform correct learning. In contrast, in heterogeneous images, there are significant geometric deformations and nonlinear radiometric differences between optical and SAR images, which make large differences between the acquired score maps of these images. As shown in Fig. 5(b), when the local range is 8×8, there is a large difference in the location of the maximum score value in the same area, and the maximum score value is also different. When the local range is 16×16, the local maximum scoring points A_1 and B_1 are obviously not the same location points, so when the location optimization is performed, this pair of different location points will be considered as the same name points for optimization, which makes the network perform incorrect learning.

Because of the large difference in the scoring values of the same positions of optical and SAR images, it is difficult to optimize the scoring points of the same positions by constraining the scoring values. There are some opportunities to address this issue with the position-based constraint. Inspired by the window-based loss for keypoint detection in [29], we design a position-consistent loss function for the keypoint detection of optical and SAR images. We design a selective judgment loss, which reduces the number of point pairs where the loss function is incorrectly calculated and prevents backpropagation from causing the network to perform incorrect learning. The detailed procedure of our designed loss function is given as follows.

In the score map $S \in \mathbb{R}^{H\times W}$, the score values of $W_{(u,v)}^{(i)}$ are exponentially scaled within each $N \times N$ window and normalized according to the following formula:

$$M_{(u,v)}^{(i)} = \frac{e^{W_{(u,v)}^{(i)}}}{\sum_{j=0}^{N}\sum_{k=0}^{N} e^{W_{(j,k)}^{(i)}}} \tag{15}$$

where $M_{(u,v)}^{(i)}$ is the result of $W_{(u,v)}^{(i)}$ after exponential transformation and normalization, (u, v) denotes the coordinates within the $N \times N$ window, and $W^{(i)}$ denotes the ith $N \times N$ size window. After the index transformation, the maximum scored value within the local window becomes the primary position. By multiplying the index value of the window, we can find the index coordinates of the local maximum within its

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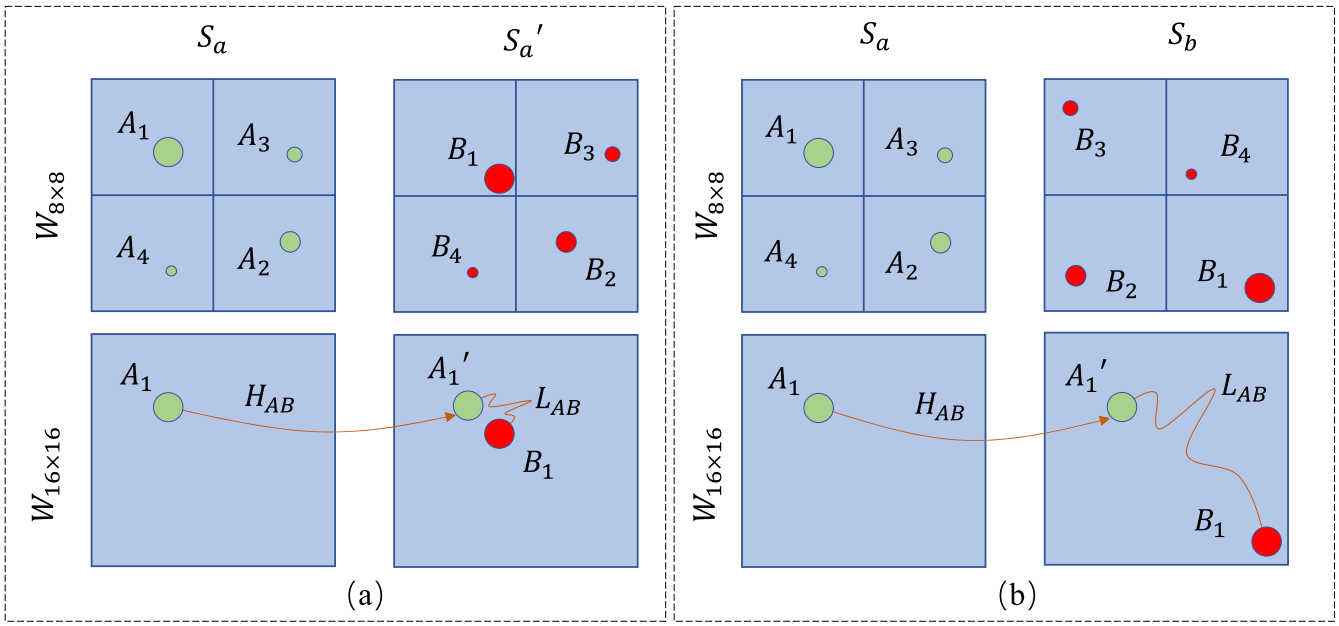


Fig. 5. Relationship between scores of homologous and heterologous images. (a) Score relationship of homologous images. (b) Score relationship of heterologous images. S_a and S_a' represent the optical image score map, S_b represents the SAR image score map, and $W_{8\times8}$ and $W_{16\times16}$ indicate the size of the local range. The circle size in the image represents the score value. A_1, A_2, A_3, and A_4 and B_1, B_2, B_3, and B_4 represent the score values, and $A_1 > A_2 > A_3 > A_4$ and $B_1 > B_2 > B_3 > B_4$. H_{AB} represents the geometric transformation relationship of the images, and L_{AB} represents the distance between A_1 after geometric transformation and B_1.

window $[x^{(i)}, y^{(i)}]^T$, which is calculated as follows:

$$\left[x^{(i)}, y^{(i)}\right]^T = \sum_{[u,v]\in W^{(i)}} M^{(i)}_{(u,v)}[u, v]^T + C_{W^{(i)}} \tag{16}$$

where $C_{W^{(i)}}$ denotes the coordinates of the upper left corner of the $W^{(i)}$ window. From (15) and (16), we can obtain the coordinates of the local maximum score value within the window, and the obtained keypoint coordinates are differentiable and can be backpropagated well. For two given score maps S_a and S_b, the distance between their local maximum score points within the corresponding windows is calculated as follows:

$$L_{(x^i_{ab}, x^i_{ab})} = \left\| \left[x^{(i)}, y^{(i)}\right]^T_a - H_{ba}\left[\tilde{x}^{(i)}, \tilde{y}^{(i)}\right]^T_b \right\|^2 \tag{17}$$

where H_{ba} is the score map S_b to S_a geometric transformation relation and $[\tilde{x}^{(i)}, \tilde{y}^{(i)}]$ is the actual maximum coordinate obtained by nonmaximum suppression.

Due to the discrepancy between the optical and SAR image score maps, we need to filter out the wrong point pairs before calculating the loss to avoid affecting the backpropagation process of the network, which leads to incorrect learning. For this purpose, we constrain the loss function by using the position difference of the maximum scored points within the corresponding window as a condition, and our conditional constraint is given as follows:

$$L_{(x^i_{ab}, x^i_{ab})} = \begin{cases} 0, & \text{if } L_{(x^i_{ab}, x^i_{ab})} > D \\ L_{(x^i_{ab}, x^i_{ab})}, & \text{otherwise} \end{cases} \tag{18}$$

where D is the distance threshold between the two points, and we will discuss the effect of its value on the detection results of the ablation experiment in Section IV. When the distance between the positions of the two maximum scored points within the corresponding window exceeds D, we consider that the true positions of the pair of scored points are not consistent and are wrong pairs of points. Therefore, we keep them from the loss calculation by assigning the distance between them as 0 to prevent backpropagation from influencing the network to learn correctly when backpropagating. Then, the loss computation between the score maps S_a and S_b with a window of size $N \times N$ is given as follows:

$$\begin{aligned} L_{ab}(S_a, S_b, H, N) &= \sum_i \alpha^{(i)} L_{(x^i_{ab}, x^i_{ab})} \\ \alpha^{(i)} &= R^a\left[x^{(i)}, y^{(i)}\right]_a + R^b\left[\tilde{x}^{(i)}, \tilde{y}^{(i)}\right]_b \end{aligned} \tag{19}$$

where $\alpha^{(i)}$ is the weight based on the score plot and R^a and R^b are the response plots of S_a and S_b, respectively, with coordinates related to a single response matrix.

In heterogeneous image keypoint detection, the use of too many windows does not bring better detection results, so we use a window size of [8,16,32], and the window weighting factor is [16,4,1]. We optimize the position of the maximum score point between the two score maps with a two-way constraint. The losses between score maps S_a to S_b and score maps S_b to S_a are calculated and averaged as the final loss with the following equation:

$$L_d = \sum_l \gamma_l((L_{ab}(S_a, S_b, H, N_l) + L_{ba}(S_b, S_a, H, N_l))/2) \tag{20}$$

where γ_l is the weight factor of the window and l is the size of the window.

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IV. Experiments

In this section, we present the data used for the experiments, implementation details, evaluation methods, comparison of keypoint detection experiments, application of keypoint detection in matching, and ablation experiments.

A. Experimental Data

1) Training Data: The training data are obtained from the optical and SAR datasets in MU-Net [55]. This dataset contains 512 × 512 pixel precisely aligned optical and SAR image pairs covering different scenes, such as towns, villages, mountains, farmlands, rivers, and forests. This dataset was acquired in May 2021 by Sentinel-1 and Sentinel-2 with a resolution of 10 m. We randomly rotate the images [−45°, 45°] and crop them into 256 × 256 images by their centers, and we use the corresponding regions of the images as our training set. For the geometrically transformed optical and SAR image pairs, we generate binary masks based on their geometric transformation relationships. The common regions between the images are extracted by the binary mask to avoid nonexistent regions for keypoint detection. We use optical and SAR images of size 256 × 256 after random geometric transformation as the training data.

2) Test Data: To ensure the general applicability of our keypoint detection method, we divide the test data into two categories: test datasets 1 and 2, which are from different datasets and have different geometric transformation forms. The data in test dataset 1 are from the same optical and SAR datasets in MU-Net [55] as the training data, and the test data were not used for training. We divide this dataset into five types of scenes: towns, suburbs, mountains, rivers, and forests. Ten pairs of images are randomly selected for each type of scene, and the images are 512 × 512 pixel precisely aligned image pairs. The data in data 2 come from the WHU-SEN-City [56] dataset and contain precisely aligned optical and SAR image pairs covering 32 cities in China. This dataset includes different scenes, such as mountains, forests, lakes, rivers, roads, bridges, farmlands, and buildings. This dataset was obtained from Sentinel-1 and Sentinel-2 with a spatial resolution of 10 m. We crop the images into 512 × 512 size image pairs and rotate them randomly [−45°, 45°]. As in dataset 1, we divide dataset 2 into five categories of scenes, towns, suburbs, mountains, rivers, and forests, and ten pairs of images are randomly selected for each category. The training data and test data are shown in Fig. 6.

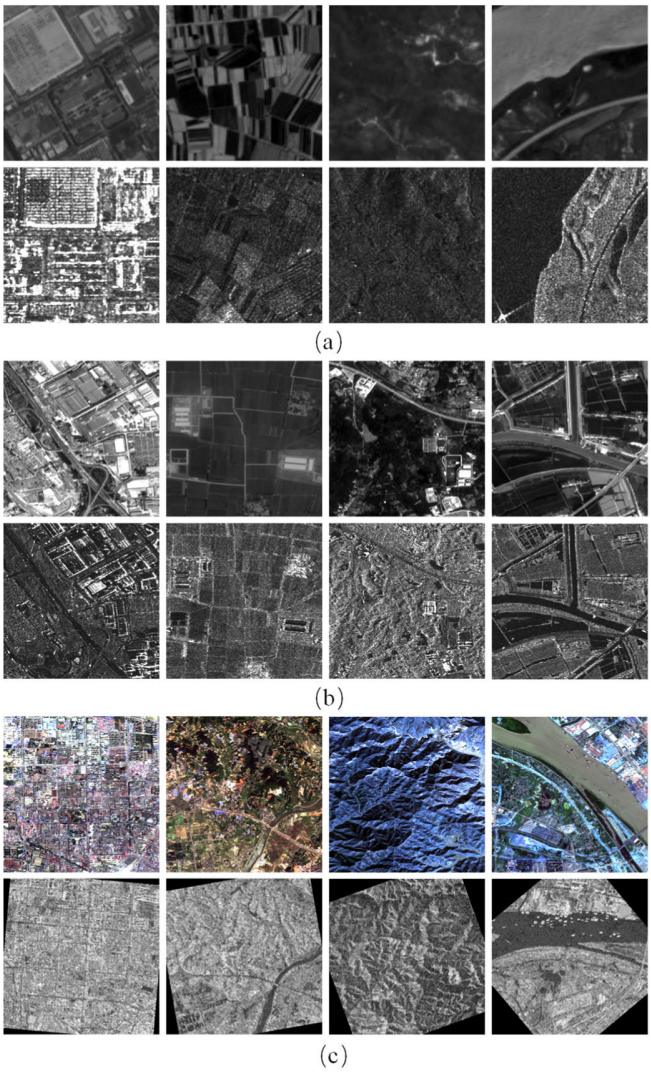


Fig. 6. Some examples of the training dataset and test dataset. (a) Training dataset, (b) test dataset one, and (c) test dataset two. The first row image is an optical image, and the second row image is an SAR image. The SAR image in test dataset 2 is shown as a grayscale map.

B. Implementation Details

Our keypoint detection method is implemented in PyTorch and trained by random geometric transformations to obtain images of size 256 × 256. The network is trained on an NVIDIA RTX3060 GPU and requires 30 iterations per training, with a batch processing size of 16. The last round of each training iteration is weighted with parameters that will be used in the final model for this training. It has an initial learning rate of 0.01 and a dynamically updated learning rate for five training rounds per cycle, optimized using the Adam optimizer. For testing, we upsample and downsample the original image to extract features at different scales with a scale factor of $\sqrt{2}$, upsampling once and downsampling four times. The keypoints detected at different scales are used as our final detection results.

C. Experimental Evaluation Indicators

1) Repeatability: We use the evaluation method proposed in [57] and further improved in [20] and [58]. The repetition rate is the ratio between the number of homonymous points detected and the total number of keypoints detected for both images. For a pair of images, two keypoints are homonymous if their coordinates satisfy the following:

$$\left|P_i(x, y)H_{ij} - P_j(x, y)\right| < d \tag{21}$$

where $P_i(x, y)$ is the coordinate of the ith keypoint of the first image, $P_j(x, y)$ is the coordinate of the jth keypoint of the second image, H_{ij} is the transformation relationship from the first image to the second image, and d is the distance threshold.

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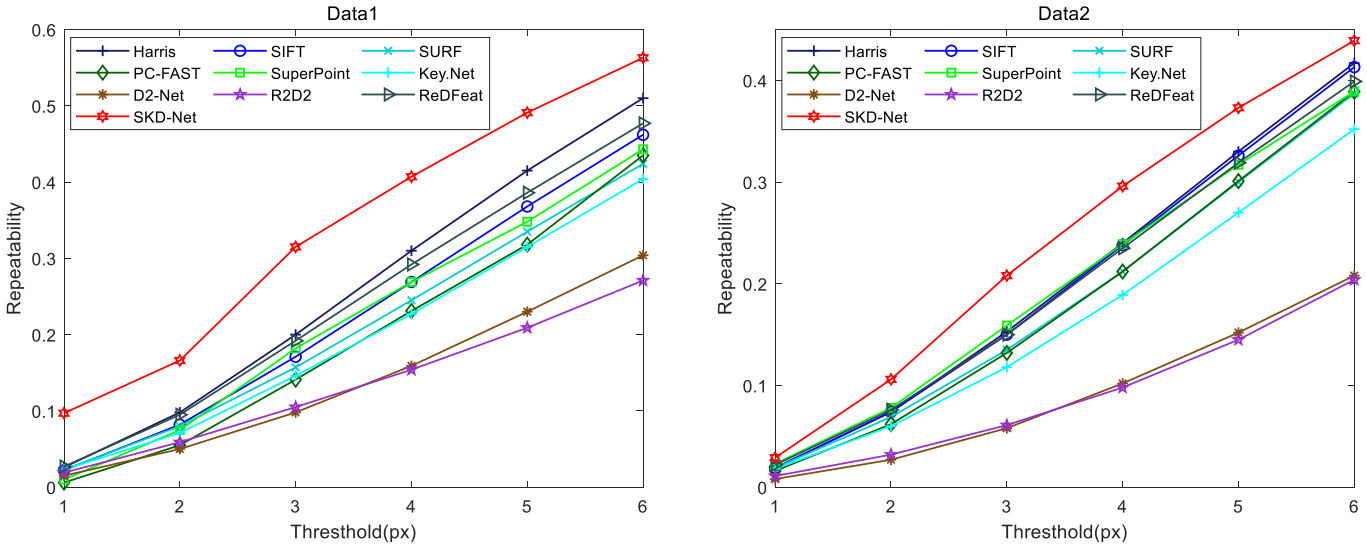


Fig. 7. (Left graph) Average repeatability rate of each detection method on dataset 1 with pixel thresholds from 1 to 6. (Right graph) Average repeatability rate of each detection method on dataset 2 with pixel thresholds from 1 to 6.

To compare different detection methods, we fix the number of extracted keypoints. Too many points will unnecessarily increase the computation time, and too few points will lead to unsuccessful image matching, according to [19], in which the total number of keypoints required was determined to be 0.4% of the input image size. Since the size of our test image is 512 × 512, we fix the number of keypoints extracted when we compare different detection methods to 1000.

2) Evaluation Metrics for Image Matching: To evaluate whether the keypoints we detect are valuable in matching, we introduce four evaluation metrics, correct match points [29], precision [57], recall [59], and $F1$-measure [60], to evaluate the goodness of image matching. The number of correct matching points is the number of correct point pairs of the matching results. Precision is the ratio of the number of correct matching point pairs of matching results to the sum of the number of correct and incorrect matching point pairs of the matching results. Recall is the ratio of the number of correct matching point pairs of the matching result to the total number of true homonymous point pairs in the initial matching point set. The $F1$-measure captures the applicability between true homonymous points and detected correct matching point pairs by jointly considering recall and precision and is calculated as

$$F1 = (2 * \text{Pre} * \text{Rec})/(\text{Pre} + \text{Rec}) \tag{22}$$

where $F1$ is the $F1$-measure, Pre is the precision, and Rec is the recall.

D. Keypoint Detection

In this section, we present the Harris [12], SIFT [14], PC-FAST [16], SURF [15], SuperPoint [28], Key.Net [29], D2-Net [34], R2D2 [27], and ReDFeat [37] detection methods and our proposed multilevel attention Siamese network for keypoint detection in optical and SAR images (SKD-Net) detection method experimental results.

We measure the average repetition rate of each detection method with test datasets 1 and 2 at pixel thresholds of 1–6 to better show their true performance. As shown in Fig. 7, SKD-Net has significantly higher repetition rates than those of the other detection methods at different pixel thresholds. In dataset 1, SKD-Net is approximately 36% better than that of the detection performance of the other methods, and in dataset 2, SKD-Net is approximately 20% better than that of the detection performance of the other methods. The repetition rate of each detection method in dataset 2 is generally lower than that of dataset 1, mainly because the local feature variability of the image becomes larger after the rotational transformation in dataset 2, and there are fewer areas of repetition between the optical and SAR images.

To show the performance of each detection method in more detail, we measure the repetition rate of each detection method in different scenes of test datasets 1 and 2 (with a pixel threshold of three). We extract 1000 keypoints for each detector separately, and since the SuperPoint [28] detector cannot extract exactly 1000 keypoints, we made the number of keypoints extracted close to 1000 by adjusting the threshold value.

As shown in Table I, SKD-Net has a significantly higher repetition rate than that of the other detectors for different scenarios with different data. The highest obtained value is up to 18.25% higher than that of the repetition rate of the other detectors, and the lowest obtained value is up to 3.43% higher than that of the repetition rate of the other detectors. From the table, we can see that D2-Net [34] and R2D2 [27] have lower repetition rates in different scenes with different datasets, mainly because they are used in homogenous images and are very sensitive to nonlinear radiation, and keypoint detection is strongly influenced by the descriptions, which are not applicable in optical and SAR image keypoint detection. Apart from our detectors, Harris [12], SuperPoint [28], and ReDFeat [37], correspondingly higher repetition rates are

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TABLE I

AVERAGE REPEATABILITY RATES OF DIFFERENT DETECTION METHODS UNDER DIFFERENT SCENARIOS FOR TEST DATASETS 1 AND 2, THE PIXEL THRESHOLD IS THREE

Method	Repeatability (%)									
	Data1					Data2				
	Urban	Suburban	Mountain	River	Forest	Urban	Suburban	Mountain	River	Forest
Harris	17.21	22.98	20.85	19.04	21.23	14.19	15.57	17.21	14.37	15.13
SIFT	15.38	18.44	17.22	17.42	18.26	14.43	15.76	16.49	13.44	15.83
SURF	13.98	18.33	14.14	16.37	14.28	14.19	13.53	14.71	11.60	13.81
PC-FAST	13.48	15.13	13.37	14.29	14.01	11.69	13.17	14.74	13.03	13.5
SuperPoint	16.59	22.49	15.88	17.43	16.04	15.50	14.98	17.94	14.99	15.12
Key.Net	13.64	19.13	13.24	11.46	14.27	12.03	13.15	13.09	15.68	12.87
D2-Net	9.99	8.83	9.89	10.46	10.60	5.84	5.29	6.64	5.62	4.82
R2D2	8.68	16.51	10.75	6.05	10.59	6.70	6.05	6.29	5.19	5.22
ReDFeat	18.51	20.73	18.06	19.49	19.03	14.07	15.59	15.23	15.04	15.47
SKD-Net	**23.82**	**41.23**	**31.47**	**29.58**	**35.0**	**18.93**	**20.31**	**24.34**	**19.49**	**19.86**

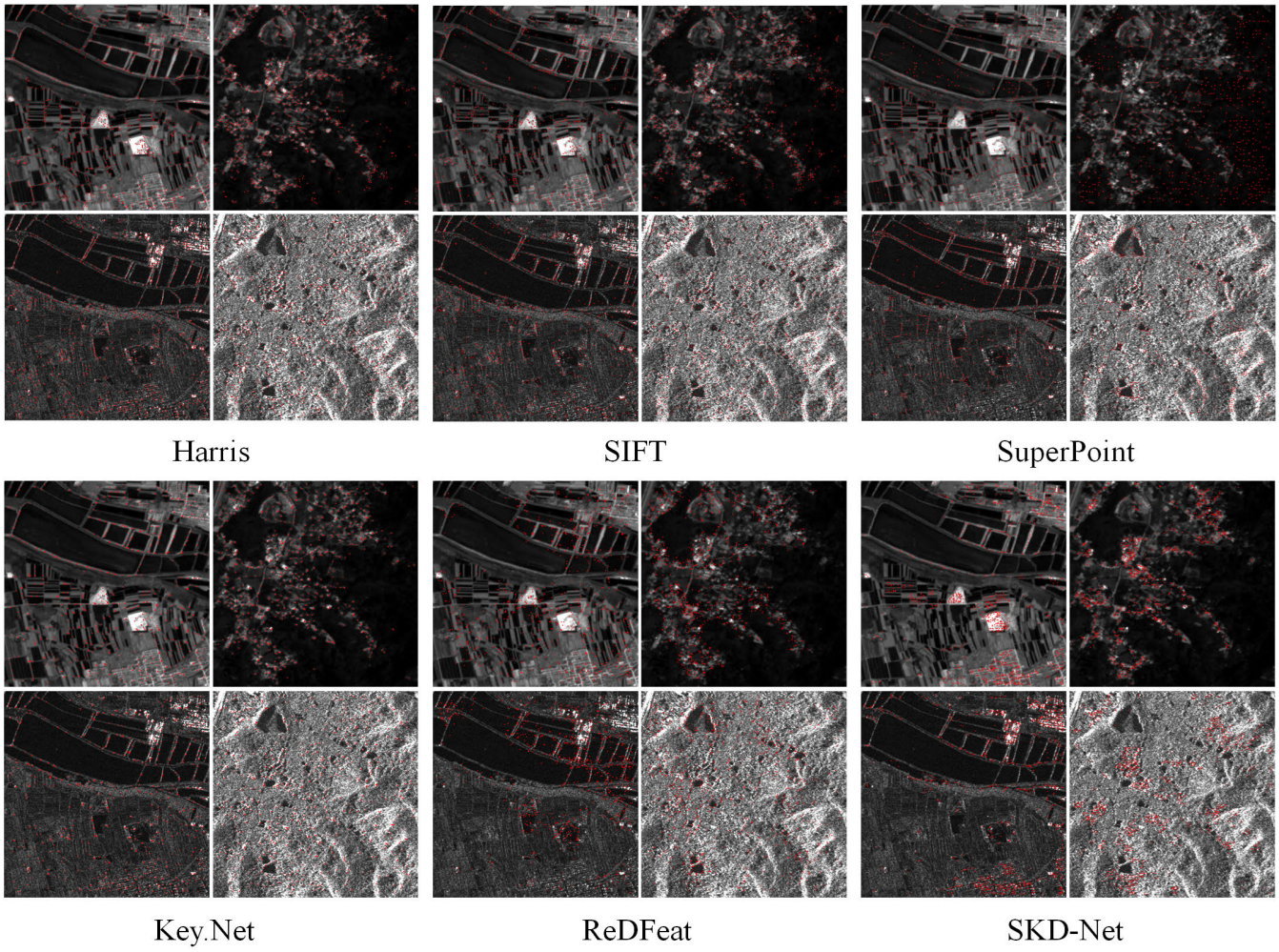


Fig. 8. Keypoint detection results of Harris, SIFT, SuperPoint, Key.Net, ReDFeat, and SKD-Net are shown in test dataset 1. (Top row) Each method represents optical images. (Bottom row) SAR images.

obtained in different data scenarios. Because the optical and SAR image data we use have more structural information, the Harris [12] detector and SuperPoint [28] detector, which rely on the detection of corner point information, can obtain higher repetition rates, while the ReDFeat [37] detection method for heterogeneous images does not have significantly better results than that of the homogenous detector. This is mainly because ReDFeat [37] constrains each other by detection

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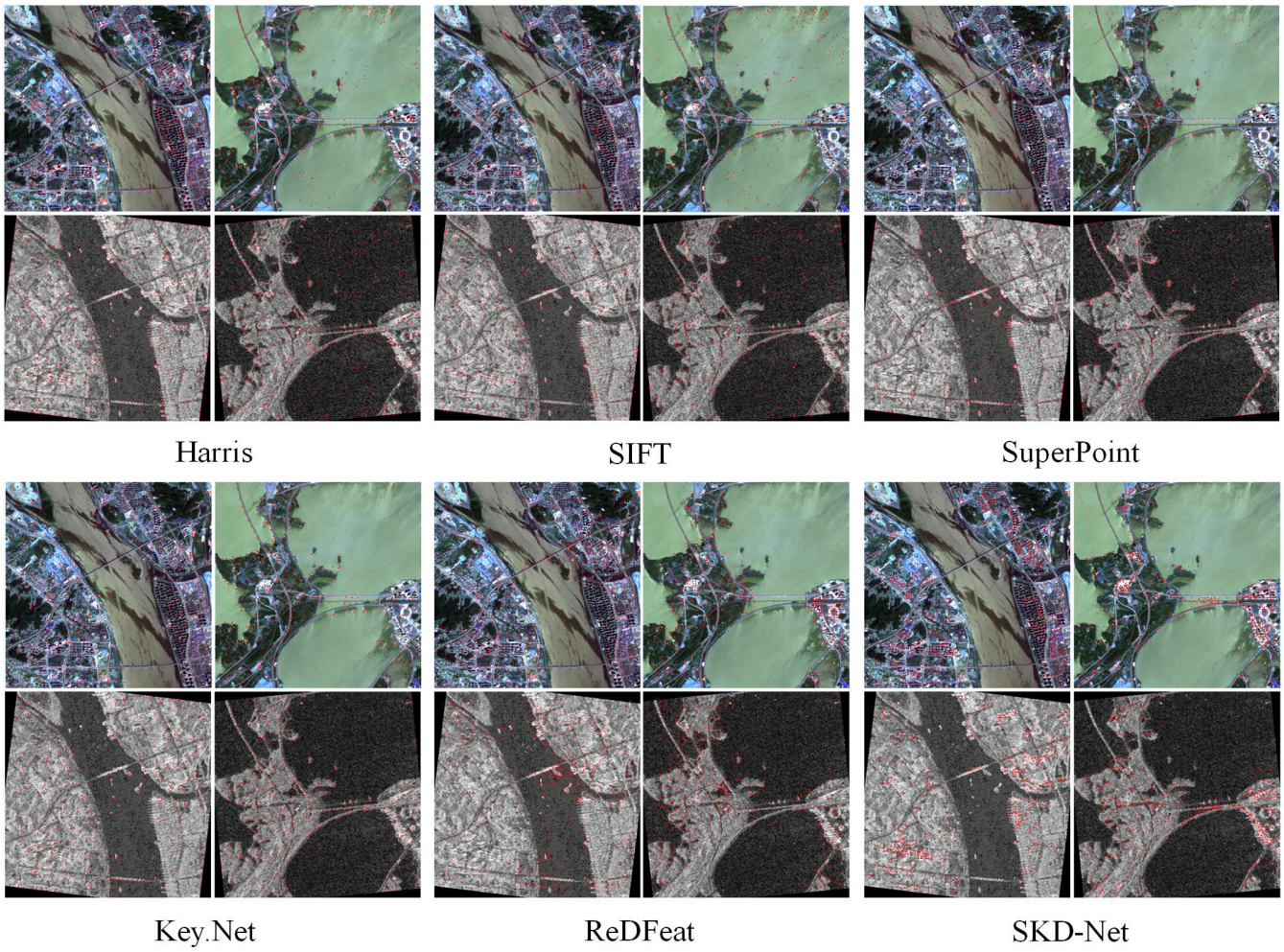


Fig. 9. Harris, SIFT, SuperPoint, Key.Net, ReDFeat, and SKD-Net keypoint detection results are shown in test dataset 2. (Top row) Each method is the optical image. (Bottom row) SAR image.

and description during the training process, and the network focuses more on the description part. In the network design, ReDFeat [37] first gets the descriptor features and then obtains the score map of keypoint detection based on the descriptor features. This strengthens the connection between descriptors and keypoint detection to some extent, but at the same time, the keypoint detection part is strongly influenced by descriptors, so it does not obtain a high repetition rate in the keypoint detection part. In contrast, our method only focuses on the keypoint detection part, and the keypoint detection is not constrained by the descriptors.

When simply evaluating the index of the repetition rate, we are unable to see the location and distribution of keypoints. To compare the performance of the detection methods more intuitively, we select several detectors with high repetition rates (Harris [12], SIFT [14], SuperPoint [28], Key.Net [29], and ReDFeat [37]) and SKD-Net for comparison. We randomly select two pairs of images in test datasets 1 and 2 for the keypoint detection demonstration.

As shown in Figs. 8 and 9, in terms of distribution, SKD-Net is mainly concentrated in regions with more complex texture than ReDFeat [37]. However, in terms of location, SKD-Net clearly has better extracted keypoint locations than ReDFeat [37], and our keypoints basically fall in regions with features, while a small number of keypoints extracted by ReDFeat [37] will appear in regions without texture. Unlike Harris [12], SIFT [14], SuperPoint [28], and Key.Net [29], our method focuses more on regions with complex textures, and more keypoints are extracted in regions with complex textures, while relatively fewer keypoints are extracted in regions with weaker textures. Although the other methods detect a more uniform distribution of keypoints, when they extract more keypoints, they appear in regions without texture, which is not conducive to the subsequent description for matching work.

E. Image Matching

Although we have demonstrated that SKD-Net has better results from the qualitative and quantitative evaluations, the keypoints of our detection are for optical and SAR image matching. To demonstrate that the keypoints of SKD-Net detection are favorable for matching, we will further verify them in image matching. To compare the performance of each detection method in image matching more fairly, we combine each detector and the same descriptor for image matching for test datasets 1 and 2. We choose the traditional descriptor

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TABLE II

MATCHING PERFORMANCE OF VARIOUS DETECTORS AND DESCRIPTORS, INCLUDING HODM, HYNET ADD HARDNET, WITH TEST DATASETS 1 AND 2. NUM INDICATES THE NUMBER OF CORRECT MATCHES, PRE INDICATES THE PRECISION, REC INDICATES THE RECALL, AND $F1$ INDICATES THE $F1$-MEASURE

Method	Data1				Data2			
	Num	Pre	Rec	F1	Num	Pre	Rec	F1
Harris+HoDM	91.8	0.105	0.46	0.169	72.2	0.083	0.432	0.139
SIFT+HoDM	81.6	0.092	0.475	0.153	73.8	0.085	0.451	0.142
SURF+HoDM	93.6	0.097	0.56	0.165	66.6	0.069	0.472	0.121
PC-FAST+HoDM	67.9	0.078	0.499	0.134	62.7	0.071	0.43	0.121
SuperPoint+HoDM	92.1	0.104	0.485	0.169	76.8	0.087	0.435	0.143
Key.Net+HoDM	88	0.097	0.562	0.16	60.6	0.065	0.55	0.115
D2-Net+HoDM	50.1	0.056	0.541	0.101	30.6	0.035	**0.563**	0.066
R2D2+HoDM	74.9	0.084	0.545	0.142	24.1	0.027	0.481	0.051
ReDFeat+HoDM	111.3	0.114	**0.574**	0.189	85	0.088	0.548	0.145
SKD-Net+HoDM	**156**	**0.179**	0.469	**0.254**	**107.5**	**0.125**	0.43	**0.19**
Harris+HyNet	21.7	0.057	0.112	0.074	21.7	0.056	0.129	0.078
SIFT+HyNet	20.9	0.055	0.125	0.075	23.3	0.06	0.142	0.084
SURF+HyNet	23.3	0.06	0.139	0.083	16	0.041	0.115	0.06
PC-FAST+HyNet	17.2	0.045	0.125	0.066	15.9	0.042	0.111	0.06
SuperPoint+HyNet	25.1	0.068	0.129	0.087	22.7	0.06	0.131	0.081
Key.Net+HyNet	21.9	0.059	0.128	0.077	17.8	0.047	0.154	0.071
D2-Net+HyNet	14.3	0.037	0.15	0.059	11.1	0.031	**0.218**	0.053
R2D2+HyNet	20.9	0.057	**0.159**	0.08	9.1	0.025	0.164	0.042
ReDFeat+HyNet	26.4	0.070	0.136	0.092	22.9	0.059	0.153	0.084
SKD-Net+HyNet	**37.9**	**0.099**	0.118	**0.105**	**33.6**	**0.087**	0.138	**0.105**
Harris+HardNet	27.9	0.074	0.144	0.096	22.4	0.059	0.135	0.081
SIFT+HardNet	21.9	0.059	0.131	0.08	21.9	0.057	0.135	0.08
SURF+HardNet	23	0.062	0.136	0.084	15.9	0.042	0.114	0.061
PC-FAST+HardNet	18.8	0.05	0.13	0.071	18.6	0.049	0.13	0.071
SuperPoint+HardNet	27.3	0.074	0.143	0.095	22.9	0.061	0.134	0.083
Key.Net+HardNet	23.5	0.065	0.144	0.086	17.5	0.047	0.148	0.07
D2-Net+HardNet	15.3	0.041	**0.165**	0.066	10.0	0.028	**0.189**	0.048
R2D2+HardNet	21.1	0.059	0.141	0.08	8.8	0.024	0.165	0.041
ReDFeat+HardNet	28.2	0.074	0.147	0.097	19.3	0.051	0.125	0.071
SKD-Net+HardNet	**42.5**	**0.112**	0.139	**0.120**	**31.7**	**0.083**	0.132	**0.1**

HoDM [61] and the deep learning descriptors HyNet [62] and HardNet [63]. All methods match and filter features with the mutual nearest neighbor (MNN) strategy. The numbers of correct matching points, precision, recall, and $F1$-measure are calculated as the average of the test dataset with a 3-pixel threshold error.

As shown in Table II, because of the high repetition rate of the detected keypoints and the high distinguishability among the detected points, the numbers of correctly matched points, precision values, and $F1$-measure for test datasets 1 and 2 when SKD-Net is combined with the HoDM [61], HyNet [62], and HardNet [63] descriptors for matching achieves better results. The relatively low recall is mainly due to the high number of homonymous points detected by SKD-Net and the relatively low occurrence value when calculating the recall. However, D2-Net [34], R2D2 [27], and ReDFeat [37] perform better in terms of recall with different data because they are joint detection and description of keypoint detection methods, and the detected keypoints are more conducive to description. However, they perform poorly in terms of the number of correctly matched points, precision, and $F1$-measure evaluation index and do not meet the matching requirements. In addition, in Table II, we can see that the combination of each detector and the HyNet [62] and HardNet [63] deep learning descriptors for matching is not combined with the traditional HoDM [61] descriptor mainly because the HyNet [62] and HardNet [63] descriptors are used in homologation, are very sensitive to nonlinear radiation differences, and are not suitable for optical and SAR image descriptions. However, this does not affect the fairness of our comparison with other detectors. In conclusion, our detected keypoints are more distinguishable, better for image matching, and can be combined with traditional descriptors and deep learning descriptors well.

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TABLE III
AVERAGE REPEATABILITY RATE OF KEYPOINT DETECTION FOR DIFFERENT VALUES OF VOCABULARY LENGTH L IN THE TRANSFORMER MODULE

L	2	4	6	8
Repeatability (%)	27.19	**31.94**	28.05	30.04

TABLE IV
AVERAGE REPEATABILITY RATE OF KEYPOINT FOR DIFFERENT VALUES OF THE CONSTRAINT DISTANCE D IN THE LOSS FUNCTION

D	4	6	8	10
Repeatability (%)	31.16	**32.85**	31.22	31.17

TABLE V
AVERAGE REPEATABILITY RATE OF KEYPOINT DETECTION WITH SHALLOW AND DEEP FEATURE FUSION UNDER DIFFERENT WEIGHT COEFFICIENTS

f_1	f_2	f_3	Repeatability (%)
1	0	0	22.35
0.6	0.3	0.1	**31.70**
0.6	0.1	0.3	31.28
0.3	0.6	0.1	30.89
0.3	0.1	0.6	28.67
0.1	0.6	0.3	28.16
0.1	0.3	0.6	28.86
0.33	0.33	0.33	29.39
0	1	0	23.50
0	0	1	28.52

TABLE VI
REPEATABILITY RATE OF DETECTION METHODS BEFORE AND AFTER ADDING THE TRANSFORMER MODULE. THE NUMBERS IN THE TABLE INDICATE THE NUMBER OF INDIVIDUAL TRAININGS

Transformer	Repeatability (%)					
	1	2	3	4	5	Mean
×	25.51	25.97	26.03	25.91	25.40	25.76
√	**32.05**	**31.47**	**31.43**	**31.81**	**32.88**	**31.93**

F. Ablation Experiment

1) Parameter Discussion: To explore the effect of the length L of the set of words in the transformer module and the conditional constraint distance D in the loss function, we analyze the detection performance with different parameter values. We fix the other variables and explore the repetition rate of keypoint detection when different parameter values are used. From Table III, we can see that the repetition rate of the model is improved when the length L of the set of words is changed from 8 to 4. This indicates that a compact set of words can represent points of changing interest and that a redundant set of word lengths can hinder the performance of the model. We also observe that the repetition rate decreases when the set of words length decreases from 4 to 2. This is because the model loses information related to changes when the set of word length is too small. Therefore, we set the set of words length L to 4. As shown in Table IV, the repetition rate of the model increases gradually from 10 to 4 when the conditional constraint distance is from 10 to 4. This indicates that the distance is too large to constrain the correct keypoint pairs, and a smaller distance can act as a constraint to improve the repetition rate of keypoint detection. However, when the distance is too small, the number of constrained keypoints decreases and hinders the performance of the model, so we set the conditional constraint distance to six.

2) Multilevel Feature Fusion Coefficient: To explore the best detection performance when shallow and deep features are fused, we set different weight coefficients for the features extracted at different levels. According to the division of weight coefficients used in [36], we add a set of (0.33, 0.33, 0.33) weight coefficients. In our experiments, we analyze the performance of keypoint detection with the combination of different weighting parameters, and the specific parameter combinations are shown in Table V. We detect the average repetition rate of keypoint detection for different levels of features with the same pixel threshold and different fusion weight coefficients. From Table V, we can see that the repetition rate is higher when the weight coefficients of f_1, f_2, and f_3 are all used because more detailed local common information can be obtained from the fusion of shallow and deep features, which is more favorable for heterogeneous image keypoint detection. We also observe that the repetition rate is highest when the coefficients are in a decreasing trend because the shallow features are rich in more robust shape, line, and corner information and have a greater impact on keypoint detection. Therefore, we use feature fusion weight coefficients of (0.6, 0.3, 0.1).

3) Impact of the Transformer Module: To test the effect of the transformer module on keypoint detection, we compare the detection results with and without the transformer module. To avoid destroying the network structure, we replace the transformer module with the same convolutional module as convolution module 2 to obtain the detection results without adding the transformer module. These models are trained separately five times to measure their average repetition rate. From Table VI, we can observe that adding the transformer module increases the average repetition rate by 6.17% compared to the average repetition rate without the transformer module. This is because transformers have a more powerful feature extraction ability compared with that of a traditional CNN. Thus, the transformer module can obtain more spatial and rich semantic information, prevents the loss of local transformation information, and is more conducive to keypoint detection. To test the impact of adding the transformer module on the time cost of the network, we compared the time consumed by the network training before and after adding the transformer module. We trained the network for three rounds separately and measured the time consumed for each round of training. Through Table VII, we can see that adding the transformer module will take more time than the normal convolution

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TABLE VII
TIME COST OF DETECTION METHODS BEFORE AND AFTER ADDING THE TRANSFORMER MODULE

Transformer	Time(s)			
	1	2	3	Mean
×	30.0	24.21	23.94	26.05
√	31.33	25.24	25.14	27.24

module, but the difference in time cost between them is not significant. The reason for this is mainly that we designed the transformer module with one number of layers for TE and TD, and the parameter content is relatively small, which reduces the time consumption to some extent.

V. CONCLUSION

In this article, we propose a keypoint detector for optical and SAR images named SKD-Net, which first extracts different level features of optical and SAR images with a convolution module and a transformer module. Then, the shallow and deep features are fused to obtain local common information that facilitates keypoint detection. In terms of network optimization, we design the positional consistency loss to take the differences between the optical and SAR image score maps into account, autonomously selecting the correct keypoint pairs for optimization and constraining the network to learn correctly. We evaluate our detection method with two different datasets separately. The experimental results show that our keypoint detection method improves the detection performance by approximately 36% and 20% over that of the other methods for test datasets 1 and 2, respectively, and our detected keypoints are better in terms of the number of correctly matched points, precision, and $F1$-measure when combined with traditional descriptors and deep learning descriptors for matching. We also verify that the transformer module and multilevel feature fusion approach in the SKD-Net network are more favorable for heterogeneous image keypoint detection. In the future, we will study multimodal keypoint detection methods that are robust to geometric transformations to promote multimodal remote sensing image applications.

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